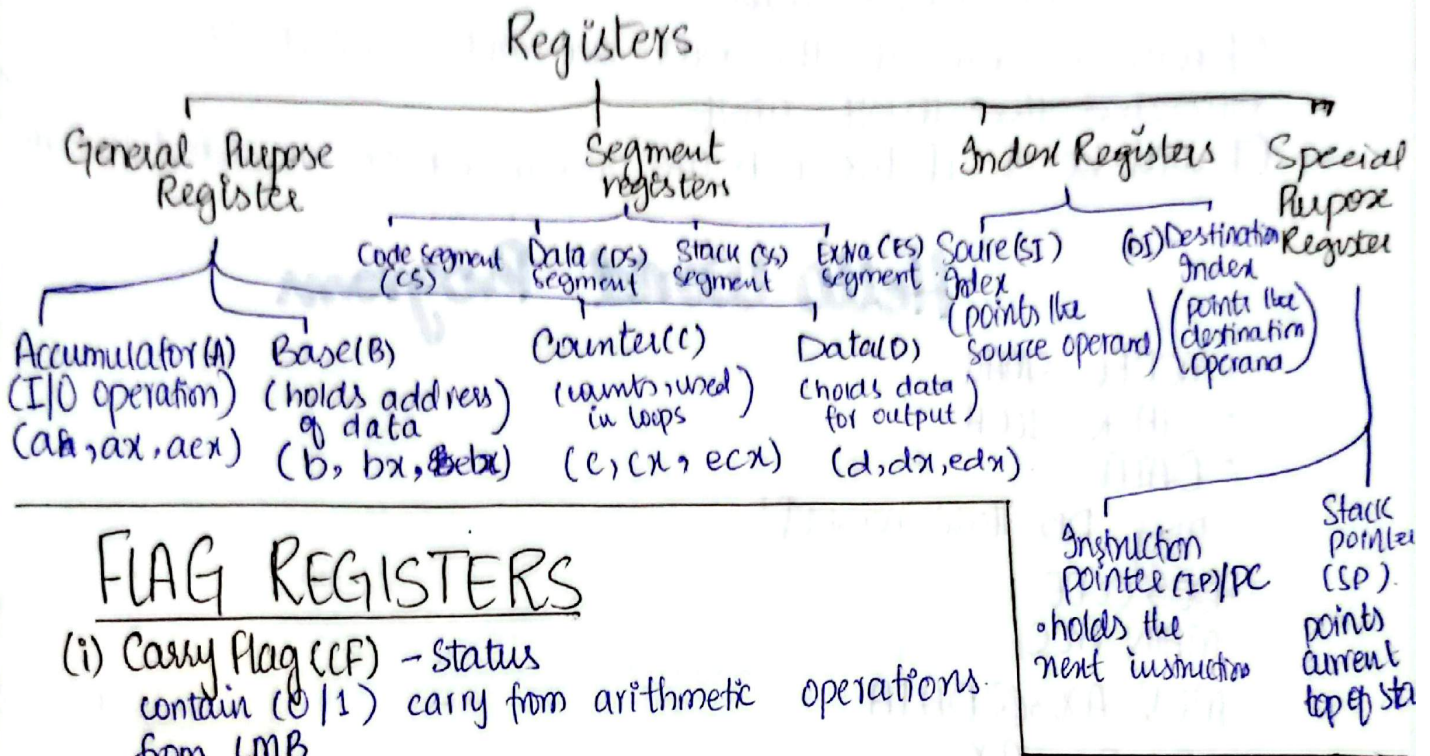


Registers

- fastest storage area
- inside CPU
- quickly accessible by CPU

X = Extended to 16 bits
E = Extended to 32 bits.



FLAG REGISTERS

- (i) **Carry Flag (CF)** - Status
contains (0/1) carry from arithmetic operations from LMB
- (ii) **Parity Flag (PF)** - Status
indicates even or odd bit parity in the received data.

Even parity	Odd parity
if 1's is even = 0	if 1's is even = 1
if 1's is odd = 1	if 1's is odd = 0
- (iii) **Auxiliary Flag (AF)** - Status
its value would be one if there is carry after the third bit
- (iv) **Zero Flag (ZF)** - Status
its value will be zero if the answer of the arithmetic operation is not zero. And its value will be 1 if the answer is 0.
- (v) **Sign Flag (SF)** - Status.
if the value of the arithmetic operation is positive then SF = 0 and if its negative then its SF = 1.
- (vi) **Trap Flag (TF)** - Control
TF = 1; activates a single step output
TF = 0; normal execution (default).
- (vii) **Interrupt Flag (IF)** - Control.
if the IF = 1; the interrupt will be processed
if the IF = 0; the Interrupt instruction will be ignored.

(viii) Direction Flag (DF) - Control

DF = 0 : text will be displayed from left to right
DF = 1 : text will be displayed from right to left.

1) Hello

(ix) Overflow Flag (OF) - Status

OF will be zero if the bits are not wasted or exceeded the 16 bit limit.

OF will be 1 if the bits are overflowed or exceeded 16 bits.

Hello World Program

```
• MODEL SMALL
• STACK 100H
• DATA
  msg DB 'Hello world$'
• DATA CODE
  MAIN PROC.
    MOV AX, @DATA
    MOV DS, AX.
    MOV DX, OFFSET msg
    MOV AH, 09h
    INT 21h
    MOV AH, 4ch
    INT 21h
    MAIN 21h
  MAIN ENDP
END MAIN.
```

loading and will be
loading and will be